Question	559
Type	MCQ
Body	The Laplace transform of $x(t)=\int_0^t e^{-3\tau}\cos 2\tau\, d\tau$ is
Option	$\frac{s-3}{(s-3)^2+4}$
Option	$\frac{s-3}{s\left((s-3)^2+4\right)}$
Option	$\frac{s+3}{(s+3)^2+4}$
Option	$\frac{s+3}{s\left((s+3)^2+4\right)}$
Correct	4
Explanation	$\cos 2t\,u(t) \rightarrow \frac{s}{s^2+4}$ $e^{-3t}\cos 2t\,u(t) \rightarrow \frac{s+3}{(s+3)^2+4}$ $\int_0^t e^{-3\tau}\cos 2\tau\, d\tau \rightarrow \frac{s+3}{s\left((s+3)^2+4\right)}$
Subject	Signals and Systems
Topic	
Correct Marks	1
Incorrect Marks	0.33
Hint	-
Video Solution	-
PYQ	
OTS ID	